

AMENDMENT NO. 1 SEPTEMBER 2024
TO
IS 15477 : 2019 ADHESIVES FOR USE WITH CERAMIC, MOSAIC AND STONE
TILES — SPECIFICATION
(First Revision)

(Page 1, clause 4.1) — Substitute the following for the existing:

‘4.1 Type 1 Adhesive

This adhesive is suitable mainly for tiles with water absorption greater than 3 percent. It is suitable for most of the clay and ceramic (non-vitrified) tiles of small dimension (size not more than 600 mm × 300 mm) and the majority of cement based backgrounds (substrates) like cement plaster, cement concrete, cement screed, etc and for interior dry area applications only.

NOTE — Type 1 adhesive generally contain varying small amounts of chemical polymers, hence the name general purpose adhesive is suitable for general, internal, clay or ceramic floor and wall tiling. The vital minor amounts of chemical polymers enable powder adhesives, which are based on bulk cement and filler, to be modified to have appropriate properties. By addition of liquid admixture to Type 1 adhesive, its properties can be enhanced to meet the performance criteria of Type 2 adhesive.’

(Page 1, clause 4.2) — Substitute the following for the existing:

‘4.2 Type 2 Adhesive

This adhesive is mainly for tiles with water absorption less than or equal to 3 percent. It is suitable for ceramic tiles, clay tiles, vitrified/fully vitrified tiles, glass mosaic tiles, all stones, dense and large dimension (up to 600 mm by 600 mm) tiles and stones (slabs), and where cement based background and location is more demanding like wet areas, submerged areas, etc. This type of adhesive is recommended for both interior floor and wall applications and for exterior floor applications. It is not suitable for external wall applications.

NOTE — By addition of liquid admixture to Type 2 adhesive, its properties can be enhanced to meet the performance criteria of Type 3 adhesive.’

(Page 2, clause 4.3) — Substitute the following for the existing:

‘4.3 Type 3 Adhesive

This adhesive is suitable mainly for tiles and stone tiles on exterior wall substrates like plaster or concrete. It is suitable for tiles like ceramic, clay tiles, Basalt tiles, vitrified, glass mosaic tiles and porcelain tiles and all-natural stone tiles.’

(Page 2, clause 4.4) — Substitute the following for the existing:

‘4.4 Type 4 Adhesive

This adhesive is suitable mainly for tiles and stone tiles which are included in above Type 1, Type 2 and Type 3 but intended for installation on dry wall board substrates. It is suitable for all types of tiles and stones except metal tiles or engineered stones to be installed on dry wall board substrates like gypsum boards, plywood, wood, calcium silicate boards, medium density fibreboards, fibre cement boards, cement boards, bison panel, etc.

For all tiles/stones of size more than 600 mm by 600 mm and up to 1 200 mm by 1 200 mm, the adhesive shall also comply with ‘S1’ category of transverse deformation. For all tiles/stones of size more than 1 200 mm by 1 200 mm, the adhesive shall also comply with ‘S2’ category of transverse deformation.

NOTES

1 Type 4 adhesive can be cementitious and non-cementitious generally.

2 For exterior drywall cladding, as agreed between buyer and purchaser, Type 4 adhesive may be of ‘S2’ category of transverse deformation.’

Price Group 1

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(Page 2, clause 4.5, Note) — Substitute the following for the existing:

‘NOTE — Type 5 adhesive is a mixture of synthetic resin, mineral fillers and organic additives in which hardening occurs by chemical reaction. They are available in one or more component forms and in essence, these cover polyurethanes, epoxy adhesives and the likes.’

(Page 2, clause 4.5) — Insert the following at the end:

‘4.6 Modification in Adhesives

The polymer modified adhesive can be achieved by the addition of a liquid admixture to the standard general purpose adhesive which is called as polymer fortified adhesive. This is a more of specialist and technical solution that provides enhanced adhesion and increased flexibility. Thus, by addition of liquid admixture to Type 1 adhesive, its properties can be enhanced to meet the performance criteria of Type 2 or Type 3 or Type 4 adhesive.

NOTE — Type 2, Type 3 and Type 4 adhesives are more technical products, generally referred as polymer modified adhesives, and are required in situations where the background (substrate) or the tile or the surface is particularly challenging.’

(Page 3, clause 5.6, line 3) — Substitute ‘category’ for ‘class’.

(Page 11, clause F-4, line 6) — Substitute ‘Category’ for ‘Class’.